

UAV Sighting Reporting System — Design Document

Oct 2, 2026 · @Thomas K

Overview

The system lets anyone in Taiwan report an illegal or suspicious drone in minutes, and routes each report to the right agency automatically. It adapts the model of Taiwan's police violation reporting portals: the reporter submits evidence with time and location, receives a case number, and later sees the outcome.

It differs from those portals in three ways. Reporters may stay anonymous. Reports are fused with sensor, registry and aircraft data into a single picture of each drone event. Cases are routed across several agencies (police, Civil Aviation Administration and defense), not one.

The system has three Flutter clients (citizen Android and web app, agency web console, field officer Android app) and an on-premise Python backend built to absorb further data sources into its analytics.

Key decisions

Every decision below was made during the design Q&A; later sections explain the consequences.

Area	Decision
What is reported	Citizen sightings of illegal or suspicious drones
Receiving agencies	Multi-agency routing: police, CAA, defense
Reporter identity	Anonymous allowed
Evidence	Photo/video, GPS + compass bearing + estimated altitude, rotor audio, Remote ID / OpenDroneID scan
Routing	Automatic, with dispatcher override

Area	Decision
Response target	Near real-time (minutes)
External data	CAA registry and permits, ADS-B, RF/radar sensors, weather, CCTV, city data
Hosting	Government cloud / on-premise in Taiwan
Analytics users	Agency dashboards and hotspot maps
Clients	Flutter: citizen Android + web, agency web console, field officer Android
Reporter feedback	Status + templated final outcome; templated evidence requests only
Scope	Full system from day one, nationwide launch
Defense cases	Same console, role-restricted
Languages	Traditional Chinese, English, Vietnamese, Indonesian, Thai, Filipino
First pickup	Each agency's own dispatch desk
Unacknowledged cases	Re-routed to a backup agency
Severity	3 levels; auto upgrade, downgrade by dispatcher
Staff login	Username + password
Retention	Everything kept indefinitely
Public visibility	Nothing public
APIs	REST + WebSocket
Backend language	Python
External feeds	Own JSON schema; vendors must deliver in it

Reporting flow and evidence capture

A report reaches a dispatcher within seconds because metadata travels first and media follows separately.

1. The reporter opens the app and taps Report. The app records GPS position, time and device attestation (Play Integrity) immediately.
2. The reporter points the phone at the drone. Compass heading gives a bearing line; phone tilt gives an elevation angle for estimating altitude.
3. In parallel, the app scans for Remote ID broadcasts via OpenDroneID and starts optional photo, video and audio capture.
4. A small metadata packet (position, bearing, elevation, Remote ID data, timestamps, file hashes) is sent at once. This triggers fusion, routing and the dispatcher alert.
5. Media uploads afterwards through resumable uploads (tus), continuing in the background via workmanager if the app closes.
6. The reporter receives a case number and a secret token.

Evidence capture details:

- **OpenDroneID:** a native Kotlin plugin wrapping the open-source OpenDroneID receiver library. The app detects which transports the phone supports (Bluetooth 4 legacy, Bluetooth 5 Long Range, Wi-Fi Beacon, Wi-Fi NAN) and tells the user. Whether Taiwan currently mandates Remote ID for the relevant drone classes needs confirming with the CAA.
- **Bearing and triangulation:** each report is a bearing line; two or more independent lines let the backend estimate the drone's real position.
- **Audio:** rotor recordings are stored from day one to train a future classifier for drone type and false-report detection.
- **Integrity:** files are hashed (SHA-256) on the device at capture, timestamped and stored unmodified, giving a chain of custody for prosecution.
- **Web reporting:** browsers cannot scan Remote ID and have unreliable compass data, so web reports are tagged lower-fidelity and the UI suggests the Android app for live sightings.

Anonymity, feedback and evidence requests

Reporters stay anonymous; credibility comes from corroboration and abuse controls, not identity.

- **Case token:** each report returns a case number plus a secret token. The token, stored hashed on the server, lets the reporter check status and answer evidence requests without an account.

- **Abuse controls:** device attestation, rate limiting per device and network, and spam scoring. Independent corroborating reports raise an incident's confidence; isolated reports from flagged devices lower it.
- **Status and outcome:** reporters see each status change and a final outcome chosen by the dispatcher from fixed templates, for example Authorized flight, Operator identified and penalty issued, Unable to locate, or Referred to another agency. Defense cases always show the generic outcome Handled by the relevant authority.
- **Evidence requests:** dispatchers can only send templated requests, such as asking for a video or another photo of the flight direction. No template ever asks a reporter to approach or follow an operator.
- **Notifications:** push via an FCM token linked to the case, never to a person. Notifications carry no case details, only a prompt to open the app.
- **Translation:** because outcomes and requests are templated, they are pre-translated into all six supported languages.

Data sources and sensor fusion

Every input becomes an Observation; the fusion engine groups observations into Incidents (one drone event); each incident becomes a Case routed to an agency.

Source	Role in the system	Confidence weight
Citizen report (Android)	Bearing line, Remote ID, media	Medium; rises with corroboration
Citizen report (web)	Location and media only	Low
Remote ID / OpenDronelD	Serial number, drone and operator position	High when present
RF / radar sensors	Continuous machine-generated tracks	High
CAA registry and permits	Matches serial to owner; marks authorized flights	Lookup, not an observation
ADS-B	Manned aircraft positions for proximity checks	Lookup, drives escalation
Weather	Visibility and wind, to weigh sighting reliability	Context

Source	Role in the system	Confidence weight
CCTV and city data	Dispatcher verification	Context

Fusion rules:

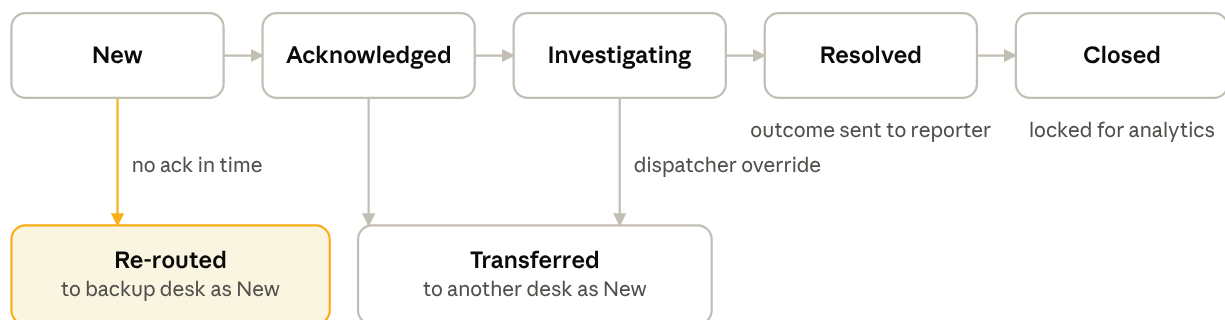
- Observations close in space and time are clustered into one incident. A surge of hundreds of reports about one drone appears to dispatchers as one incident.
- Two or more bearing lines are triangulated into an estimated position; sensor tracks and Remote ID positions override estimates.
- A Remote ID serial with a valid permit for that zone and time marks the incident Likely authorized.
- ADS-B proximity to manned aircraft escalates the incident to Critical.
- Concurrent reports about the same drone are serialized with a row lock on the incident, so a surge never creates duplicate incidents.

All external feeds must arrive in the platform's own versioned JSON schema. Vendors are contractually required to deliver it, so the schema and its conformance tests must be ready before procurement. Feeds from government partners (such as ADS-B or CAA data) are not vendors and may still need a translator owned by one side.

Routing, severity and escalation

Cases route automatically by zone to an agency desk, re-route to a backup if not acknowledged in time, and only humans can lower severity.

A case nobody acknowledges in time restarts as New at the backup desk



case lifecycle · 5 states, 2 side paths

Every transition is written as a CaseEvent. Incidents found to be the same drone are merged, keeping one surviving case.

Routing. Each Zone (geofence) defines its primary agency desk and a backup chain. Zone polygons for non-public sensitive sites exist only on the server; the citizen app shows only published CAA zones. Dispatchers can override routing by transferring a case.

Severity.

Level	Example triggers	Suggested ack timeout
Critical	Near manned aircraft (ADS-B) or an airport approach; military or critical infrastructure zone; Kinmen/Matsu restricted areas; sensor-confirmed track in a red zone	2 min
Medium	Restricted or yellow zone without a matching permit; no Remote ID in controlled airspace; hovering over residences	10 min
Low	Open area; registry match with valid permit; single unverified web report	60 min

Severity is recalculated as observations arrive. An upgrade re-alerts the assigned dispatcher and restarts the acknowledgement timer at the new, shorter timeout. A downgrade requires a dispatcher and records who and why.

Escalation.

- Routing checks a desk's on-duty roster first, so cases are not sent to unstaffed desks.
- An unacknowledged case moves to the next desk in the zone's backup chain; the original agency is notified and keeps read access.
- Every chain ends at a 24-hour national catch-all desk, such as the National Police Agency command center.
- Classified cases escalate only to cleared desks; any other backup receives a redacted case (location, time, severity).

Client applications

One Flutter monorepo (managed with Melos) produces three separate builds that share code but never ship each other's screens.

Build	Platforms	Distribution	Main features
Citizen app (<code>main_citizen.dart</code>)	Android, web	Public Play Store, public website	Report flow, OpenDroneID scan (Android), case status by token, evidence requests
Agency console (<code>main_agency.dart</code>)	Web	Government network only	Desk queues, case actions, live map, CCTV and registry lookups, dashboards
Field app (<code>main_field.dart</code>)	Android	Agency MDM / managed Google Play	Assigned cases, live drone track, operator position, on-scene Remote ID scan, officer evidence capture, offline cache

Shared packages:

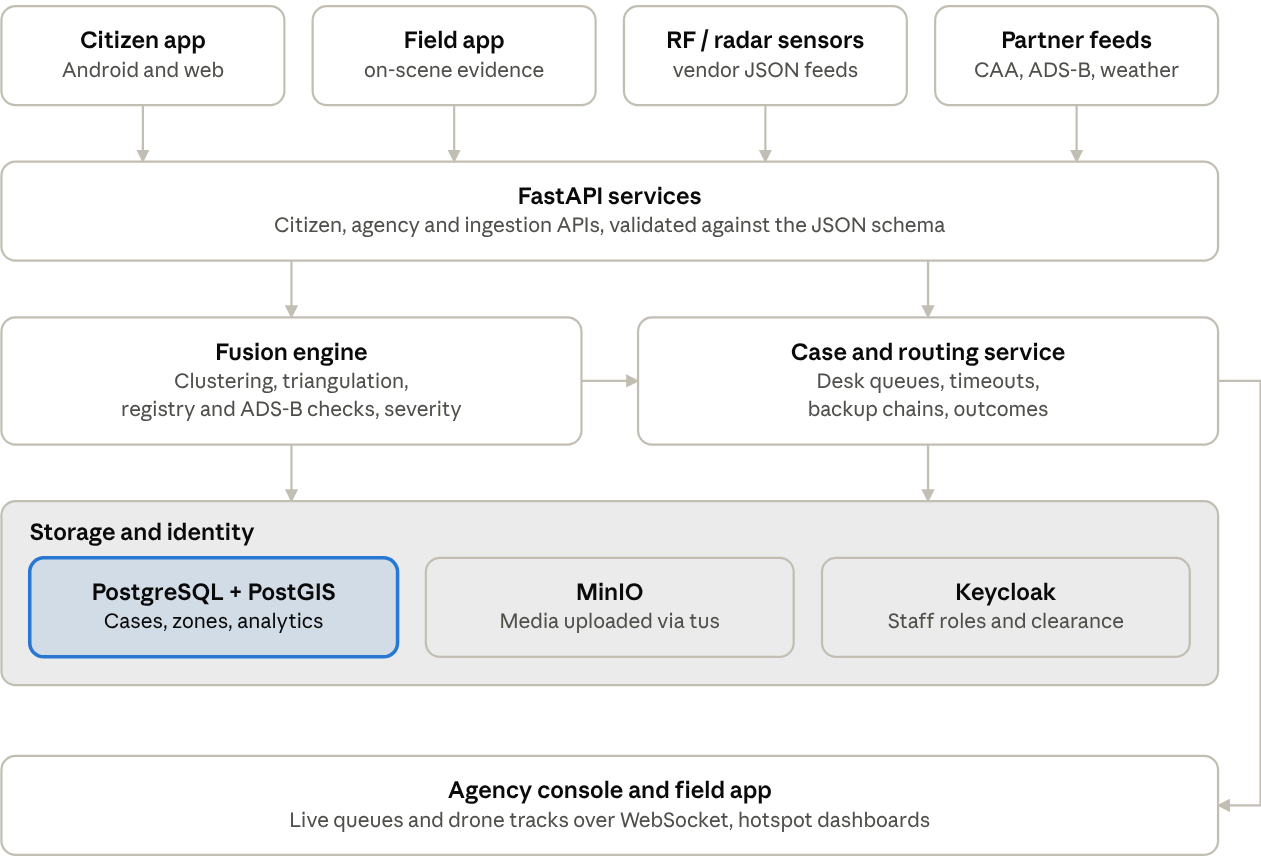
- Generated Dart API client (from the OpenAPI spec) and data models.
- Map widgets on `flutter_map` with a self-hosted tile server (OpenStreetMap or NLSC basemaps), plus zone overlays.
- Evidence capture module (camera, audio, hashing, resumable upload), reused by citizen and field apps.
- OpenDroneID native plugin (Kotlin, platform channels).
- Design system and localization: ARB files in Traditional Chinese, English, Vietnamese, Indonesian, Thai and Filipino, with bundled Noto fonts.

Defense case details appear in the field app only for defense field units; all other field users see the redacted view.

Backend architecture

The backend runs entirely on self-hostable components in government cloud or on-premise data centers in Taiwan, with PostgreSQL as the single hub: the API services read and write it directly, with no message bus.

Every input goes through one API layer straight into PostgreSQL



backend architecture · inputs to cases

All inputs enter through validated APIs; fusion turns them into incidents in PostgreSQL, and the case service drives the console and field app.

Layer	Technology	Purpose
API services	Python, FastAPI, Pydantic	Citizen, agency, ingestion and analytics APIs; OpenAPI spec generated from code
Live updates	WebSocket (FastAPI)	Console queues, severity changes, live tracks; sequence numbers for resync; PostgreSQL LISTEN/NOTIFY to reach every API instance
Background worker	Python process, same codebase	Ack timeouts, re-routing to backup desks, scheduled jobs

Layer	Technology	Purpose
Fusion engine	Python module called by the API, PostGIS queries, NumPy, Shapely	Clustering, triangulation, registry and ADS-B checks, severity
Database	PostgreSQL + PostGIS (GeoAlchemy2)	Incidents, cases, zones, users, analytics tables
Media storage	MinIO	Photos, video, audio; resumable uploads via tus
Identity	Keycloak	Agency users, roles, clearance attributes
Translation	Self-hosted machine translation	Free-text descriptions for dispatchers
Maps	Self-hosted tile server	Base maps for all clients
Platform	Kubernetes	Deployment and scaling

The fusion engine is a separate module with a narrow interface, so it can move into its own service later if load ever requires it.

Data model and APIs

Nine core entities carry the Observation, Incident, Case chain plus the organization and lookup data around it.

Entity	Purpose	Key fields
Observation	One input from any source	source type, source ID, timestamp, position or bearing line, altitude, confidence, raw payload (JSONB), schema_version
Evidence	Media attached to an observation	SHA-256 hash, media type, MinIO URI, capture time, attestation result
Incident	One fused drone event	fused track (PostGIS linestring), severity, matched zones, registry/permit match, linked observations

Entity	Purpose	Key fields
Case	Incident assigned to an agency	incident ID, agency, desk, state, assignee, outcome template, classification label
CaseEvent	Append-only audit trail	case ID, actor, action, reason, timestamp
ReporterToken	Anonymous follow-up	hashed secret, linked cases, push token, pending evidence requests
Zone	Geofence and routing rules	polygon, zone type, classification, backup chain, timeouts per severity
Agency / Desk / User / Role	Organization and access	roster, on-duty status, clearance attributes
AircraftTrack	Manned traffic from ADS-B	ICAO address, track, timestamp

Observations keep both a normalized core and the source's raw payload, so stored data can be re-processed when fusion improves. CaseEvent is never edited, only appended; it serves as both audit log and source of response-time metrics.

API groups (REST, described in one OpenAPI spec):

- **Citizen API:** submit observation metadata, resumable media upload, status by token, respond to evidence requests.
- **Agency API:** desk queues, case actions (acknowledge, transfer, merge, downgrade, resolve), registry and CCTV lookups, plus a WebSocket channel for live updates.
- **Ingestion API:** authenticated endpoints per external source, validated against the JSON schema, written to PostgreSQL.
- **Analytics API:** aggregate queries for dashboards and hotspot maps.

Security and accepted risks

Access is controlled by agency, role and clearance on every request, and two higher-risk choices are recorded here as accepted risks.

Access control:

- Attribute-based checks combine role, agency and clearance with each case's classification label.
- Field-level restriction hides sensitive fields (sensor positions, defense notes) from uncleared users, who see only a redacted case.
- Defense case data can be encrypted with keys held by the defense agency.
- Every view, export, registry lookup and CCTV access is written to an audit log.
- Lost field devices are wiped remotely through MDM.

Accepted risks:

Decision	Risk	Mitigation kept available
Staff log in with username and password only	One phished or reused password can expose defense data; security reviews under the Information Security Management Act (資通安全管理法) may reject it	Keycloak can enable TOTP or FIDO2 per role without code changes
All data kept indefinitely	Raw media holds bystanders' faces, plates and voices plus owner identities; may conflict with the Personal Data Protection Act's purpose limits; a future breach exposes all history	Storage is split into an analytics tier and an evidence tier, so a retention rule can be added to the evidence tier later

The shared console for defense cases should be reviewed by the defense side's information security officers early, before build.

Analytics and dashboards

Analytics serves agency staff only, through dashboards and hotspot maps in the agency console; nothing is published.

- **Hotspot maps:** incident heatmaps by zone, time of day and day of week, built on PostGIS.
- **Repeat offenders:** incidents grouped by Remote ID serial and by registered owner.

- **Response performance:** time to acknowledge, time to resolve, and re-route rates per agency and desk, computed from CaseEvent.
- **Source quality:** share of incidents confirmed by sensors, false-report rates by source type, Remote ID coverage.
- **Authorized versus unauthorized:** share of incidents matched to valid permits, by zone.

New data sources join analytics through the same path as live data: an ingestion endpoint, the JSON schema, and PostgreSQL. No changes to the core services are needed.

Rollout, operations and open questions

The full system launches nationwide, with each external adapter built and tested against simulated data and switched on when its agreement or hardware arrives.

Operational requirements:

- Coverage includes Taiwan, Penghu, Kinmen and Matsu; Kinmen and Matsu get the strictest severity rules.
- Every receiving agency staffs its desk 24/7, or relies on the backup chain to the national catch-all desk.
- Load tests cover genuine report surges (hundreds of reports in minutes) and spam floods disguised as surges.
- A public awareness campaign in all six languages accompanies launch.

Open questions:

- ☐ Data-sharing agreement with the CAA for registry and permit lookups
- ☐ Source of the ADS-B feed and who translates it into the platform schema
- ☐ Current Remote ID requirements in Taiwan for the relevant drone classes
- ☐ Defense information security approval for the shared console
- ☐ Which agency hosts and operates the national catch-all desk
- ☐ Legal review of indefinite retention and password-only login
- ☐ Final ack timeouts per severity, agreed with each agency